INTERIM FEDERAL SPECIFICATION

PINE OIL, DISINFECTANT (LIQUID)

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D. C., 20407, based upon currently aviilable technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address above.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers pine oil disinfectant which is used as a general disinfecting and dedorizing agent.

1.2 Classification.

1.2.1 Grade. The material covered by this specification shall be of one grade only.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

LLL-P-400a - Pine Oil.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials

ASTM Designations:

D 1544-58T - Color of transparent Liquids (Gardner Color Scale).
D 460-60 - Methods of Sampling and Chemical Analysis of Soap Products.
E 70-52T - Determination of pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

American Oil Chemists Society

A.O.C.S. Official Method F10-44 - Water Immiscible Organic Solvents

(Copies of the above method may be obtained from the American Oil Chemists' Society, 35 East Wacker Drive, Chicago 1, Illinois.)

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Official Methods of Analysis of the Official Agricultural Chemists, Post Office Box 540, Benjamin Franklin Station, Washington 4, D. C.).

3. REQUIREMENTS.

3.1 Material. The pine oil disinfectantscovered by this specification shall be a mixture of steam distilled or synthetic pine oil and soap.

3.2 Appearance. The pine oil disinfectant shall be free from sediment, suspended matter or visable evidence of separated water when tested as specified in table IV.

3.3 Odor. When tested as specified in table IV, the pine oil disinfectant shall have a pleasant mild and aromatic (piney) odor characteristic.

3.4 Chemical and physical properties. The pine oil disinfectant shall conform to the requirements as specified in table I.

Table I - Chemical and physical properties.

Requirement	Minimum	Maximum	Test paragraph
Steam-distilled or synthetic pine oil, percent by weight	80.0	--	4.5.2
Soap, percent by weight	8.0	--	Table 4
Moisture, percent by weight	--	10.0	Table 4
Phenol coefficient	5.0	--	4.5.3
Nonvolatile, percent by weight	8	11	Table 4
pH	10.0	10.6	Table 4
Viscosity, 60°F.	--	100 cps	4.5.4
Color	--	4	Table 4

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3.5 The pine oil shall conform to the requirements of table II

Table II. Quantitative requirements

Requirement	Minimum	Maximum	Test paragraph
Specific gravity at 15.5/15.5°C.	0.938	0.945	Table 4
Refractive index at 20°C.	1.479	1.492	Table 4
Dissolved water, percent by weight	--	1.0	Table 4
Acid number	--	2.0	Table 4
Distillation range, percent by volume:			
Distilled below 185°C.	--	5.0	Table 4
Distilled below 200°C.	--	25.0	Table 4
Distilled below 230°C.	95.0	--	Table 4
Total terpene alcohols, percent by weight	80.0	--	Table 4
Residue after polymerization, percent by volume	--	2.0	Table 4

3.6 Emulsion Performance. When tested in accordance with 4.5.5 the initial appearance of the emulsion shall be milky and opaque; after a 24 hour appearance the material must be stable and contain no free oil.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Supplier responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Sampling for inspection of filled unit containers. A random sample of filled unit container shall be taken by the inspector in accordance with MIL-STD-105 at inspection level I and acceptable quality level--2.5 percent defective to verify compliance with this specification in regard to fill, closure, marking, and other requirements not involving tests.

4.2.1.1 Sampling for tests. From each inspection lot (4.2.1), the inspector shall take two unit containers at random. From each of the two containers, 1-quart samples shall be taken and placed in separate, clean, dry, metal or glass containers, and then sealed, marked and forwarded to the test laboratory designated by the activity concerned.

4.3 Inspection.

4.3.1 Inspection shall be performed in accordance with Method 1011 of Fed. Test Method Std. No. 141.

4.3.1.1 Inspection of filled unit containers at the contractor's plant. Each sample filled container, selected as specified in 4.2.1, shall be examined for defects of construction of the container and closure, for evidence of leakage, and for unsatisfactory markings. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3.1.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

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Table III - Classification of Preparation for Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; illegible; improper size, location sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Buldging or distortion of container.

4.4 Lot acceptance tests. The samples selected in accordance with 4.2.1 shall be subjected to the tests specified in 4.5 except that the laboratory may perform complete tests on only one sample from each lot provided sufficient tests are made on the second sample to verify that both are alike. Lots shall be accepted or rejected by the inspector on the basis of the laboratory test report on the transmitted samples.

4.5 Test procedures.

4.5.1 The pine oil disinfectant covered by this specification shall be in accordance with the test procedures as specified in table IV and as indicated herein.

Table IV. Tests for chemical and physical requirements.

	Reference paragraph	Test methods		
		LLL-P-400a	Fed. Test Method Std. No. 141	ASTM Standards
Appearance	3.2		4261	
Pine oil, percent by weight	4.5.2			
Color	Table I			D 1544-58T
Odor	3.3		4401	
Specific gravity[1]	Table II	4.4.4		
Refractive index[1]	Table II	4.4.5	4371	
Dissolved water[1]	Table II	4.4.6	4081	
Acid number[1]	Table II		5071	
Distillation range[1]	Table II	4.4.8	4301	
Total terpene alcohol[1]	Table II	4.4.9		
Residue after polymerization percent by volume[1]	Table II		5121	
Soap, percent by weight	Table I			D 460-60
Moisture, percent by weight[2]	Table I			D 460-60
Nonvolatile, percent by weight	Table I			D 460-60
pH[3]	Table I			E 70-52

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1/ Obtain a 100 ml. sample of the disinfectant, and run a steam distillation. Set up a 500 ml. flask as the steam generator using 350-400 ml. distilled water with a glass tube running from it to within 5 mm. of the bottom of a 500 ml. flask containing the 100 ml. of sample. From the flask containing the sample, have a glass tube running to a water-cooled condenser which has a receiver at the other end to collect the condensed pine oil and water. Heat the steam generator, forcing steam to bubble through the sample and allow to run until enough pine oil has been distilled for the test. Separate the pine oil and water with a separatory funnel and finish drying the pine oil by stirring granular calcium chloride into it. Then filter the $CaCl_2$ and run the tests.

2/ Use 20 ml. of refined oleic acid per 100 ml. sample in place of sodium acetate to avoid foam.

3/ Use 50/50 dilution of disinfectant and distilled water, thoroughly stirred.

4.5.2 Pine oil (percent by weight). The percentage by weight of the pine oil (table I) shall be determined in accordance with AOCS Method F10-44, except in place of Na_2CO_3 and $CaCl_2$, step 2 of the procedure, add 5 to 10 grams of refined oleic acid, to prevent frothing. Do not use powdered or crystalline reagents as these will cause the decomposition of terpene alcohols when held at high temperatures for extended periods of time.

4.5.3 Phenol coefficient. Determine in accordance with AOAC Method 1960 Edition (amended March 15, 1964) para. 5.001 through 5.004 (using Salmonella Typhosa) (Table I).

4.5.4 Viscosity. Fill a 250 ml. or larger beaker with an undiluted sample of disinfectant. Check viscosity, using a Brookfield viscosimeter, and No. 1 Spindle, at 60 rpm (table I).

4.5.5 Emulsion performance.

4.5.5.1 Two glass stoppered 100 ml. graduated cylinders are each filled with 99 ml. of water; one with distilled water, one with 500 rpm hard water, made as follows:

0.461 gm. $CaCl_2$/Litre
0.295 gm. $MgSO_4$/Litre

To each cylinder, add an accurately measured 1 ml. sample of disinfectant. Stopper the cylinders and invert each eight times. The initial emulsion shall appear milky and opaque. After standing 24 hours at room temperature, the emulsion should not have more than 3 percent of an easily dispersible cream, and no free oil should be visible either on the surface or at the interface (table I).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.).

5.1.1 Level A. The disinfectant furnished in 1-quart, 1-gallon or 55-gallon quantities (see 6.2) shall be packaged in containers conforming to PPP-C-96, type V, class 4, oblong, PPP-D-729, type I or PPP-D-732, type I, respectively. When specified (see 6.2) a glass bottle or jug may be utilized for the 1-quart quantities.

5.1.2 Level B. The disinfectant shall be packaged as specified in 5.1.1 except plan A shall be utilized for the protective coating on the PPP-C-96 can and a commercial enamel coating is utilized on the metal drums.

5.1.3 Level C. The disinfectant shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twelve 1-quart cans or bottles (see 6.2) or 6, 1-gallon cans or jugs (see 6.2) of disinfectant shall be packed in a close-fitting box conforming to PPP-B-636, W5c. The box shall be closed and strapped in accordance with the appendix to the box specification. No packing is required for the 55-gallon drums.

5.2.2 Level B. Twelve 1-quart cans or bottles (see 6.2) or 6, 1-gallon cans or jugs (see 6.2) of disinfectant shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification. No packing required for the 55-gallon drums.

5.2.3 Level C. The disinfectant shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

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5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes.

6.1 Intended use. Pine oil disinfectant covered by this specification may be used at a 1:100 dilution for the deordorizing and disinfecting of rest rooms and general quarters. Pine oil disinfectant is only effective as a rule against gram negative organisms (i.e. S. Typhosa)

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of container required.
(c) Quantity of floor finish required.
(d) Levels of packaging and packing required (5.1 and 5.2).

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